MedFactCheck: Multi-Agent AI System for Biomedical Claim Verification

Final project - *Big Data Engineering, 2026 @ UNINA*

1 Introduction

The rapid proliferation of health-related content on the internet, social media, and even scientific preprint servers has made biomedical misinformation a critical public health challenge. Distinguishing reliable medical information from unverified or false claims requires deep domain expertise and access to up-to-date scientific evidence, a task that is increasingly difficult to perform at scale using manual processes alone.

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have opened new possibilities for automated fact-checking in the biomedical domain [2, 3]. However, existing approaches typically rely on shallow evidence sources (e.g., PubMed abstracts only) and single-model pipelines, leaving significant room for improvement in both evidence depth and reasoning transparency [7, 4].

This project asks students to build *MedFactCheck*, an end-to-end multi-agent system for automated biomedical claim verification. The system should accept claims in free-text form, sourced, for example, from social media posts, news articles, or medical web pages, and return a structured verdict (*Supported*, *Refuted*, or *Not Enough Information*) together with the scientific evidence that justifies the decision, traceable to specific passages in the literature. The core novelty with respect to prior work lies in two directions: (i) extending the evidence corpus **beyond PubMed abstracts** to include full-text articles (via PubMed Central / Europe PMC), structured knowledge bases, and other open scientific sources; and (ii) adopting a **multi-agent architecture** powered by open-source LLMs, in which specialized agents handle distinct subtasks, claim decomposition, evidence retrieval, reasoning, and verdict aggregation, rather than delegating everything to a single monolithic model.

2 Methodology

The project follows a structured pipeline divided into main phases (an example of workflow is shown in Figure 1). Students are encouraged to explore additional modules beyond those described below, provided they maintain coherence with the overall system objectives.

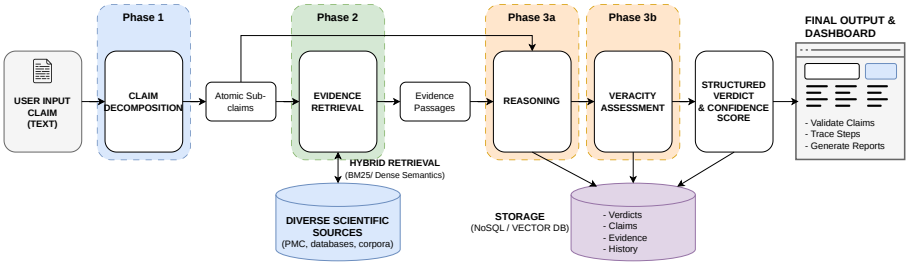

Figure 1: An example of Workflow for MedFactCheck

- **Claim Ingestion and Decomposition**:
  The first phase involves accepting an input claim, or extracting verifiable atomic claims from a longer passage, URL, or document, and decomposing complex statements into simpler, individually verifiable sub-claims. An LLM-based *Decomposer Agent* handles this step, exploiting the in-context learning capabilities of models such as LLaMA-3, Mistral, Qwen, or Gemma, available on the Hugging Face Hub[1].

- **Evidence Retrieval from Extended Scientific Sources**:
  The second phase retrieves relevant scientific evidence for each sub-claim. Crucially, the retrieval corpus must go **beyond PubMed abstracts** and incorporate at minimum one of the following extended sources:

  – Full-text articles from PubMed Central (PMC)[2] or Europe PMC[3];
  – Structured biomedical knowledge bases (e.g., UMLS, DisGeNET, UniProt);
  – Pre-processed corpora such as S2ORC or the CORD-19 dataset.

  Both sparse retrieval (BM25) and dense semantic retrieval (e.g., using BioBERT, MedCPT, or SPECTER embeddings with a FAISS index) should be explored and compared. A *Retriever Agent* orchestrates queries to the selected evidence sources.

- **Reasoning and Veracity Assessment**:
  A *Reasoning Agent* receives each sub-claim paired with the top-k retrieved evidence passages and generates a structured justification. The agent should produce chain-of-thought explanations that explicitly link evidence to the claim. A separate *Veracity Agent* (or classifier) then assigns a label (*Supported* / *Refuted* / *NEI*) and a confidence score, based on the claim text, retrieved evidence, and generated justification. Models fine-tuned on biomedical text (e.g., PubMedBERT, BioELECTRA, DeBERTa) are strong candidates for the classification step.

[1] `https://huggingface.co/models`
[2] `https://www.ncbi.nlm.nih.gov/pmc/`
[3] `https://europepmc.org/`

- **Agent Orchestration**:
  A *Coordinator Agent* manages the overall workflow, dispatching sub-claims to the appropriate agents, aggregating partial verdicts into a final decision, and handling failure cases (e.g., no evidence found). Frameworks such as AutoGen[4], LangGraph[5], CrewAI[6], or a custom implementation are all acceptable. Agent communication should be logged and inspectable.

- **Storage**:
  Claims, retrieved evidence passages, agent outputs, and final verdicts are stored in a NoSQL database (e.g., MongoDB) or a vector store (e.g., ChromaDB, Weaviate), ensuring that the full provenance chain, from input claim to source passage, is preserved and queryable.

- **Interactive Dashboard**:
  The final phase involves an intuitive interface for exploring and validating system outputs. The dashboard may be implemented using Streamlit[7], Dash[8], or similar frameworks, and should allow users to:

  - submit new claims for verification in real time;
  - inspect the verdict, justification, and source passages for each claim;
  - navigate the agent execution trace (which agents ran, in what order, with what outputs);
  - filter and search the database of previously verified claims;
  - optionally, generate a structured PDF report for a selected claim using libraries such as ReportLab[9].

Based on the methodology provided as a starting point, students are invited to explore additional modules or architectural choices, such as multi-hop retrieval, claim-level hallucination detection, or confidence calibration, while maintaining coherence with the system's goal and ensuring reproducibility.

3 Objective

The main objective is to build a scalable, transparent, and evidence-grounded system for biomedical claim verification that demonstrates how multi-agent LLM architectures can move beyond single-step retrieval-and-classify pipelines.

[4] `https://microsoft.github.io/autogen/`
[5] `https://langchain-ai.github.io/langgraph/`
[6] `https://www.crewai.com/`
[7] `https://docs.streamlit.io/`
[8] `https://dash.plotly.com/`
[9] `https://www.reportlab.com/dev/open-source-reportlab/`

3.1 Practical Examples

- **Example 1: Social Media Health Claim**
  A user submits the claim *"Vitamin D supplementation reduces the risk of COVID-19 infection"*. The Decomposer Agent identifies it as atomic. The Retriever Agent queries PubMed and PMC full texts, returning a set of RCTs and meta-analyses. The Reasoning Agent produces a chain-of-thought analysis noting that evidence is mixed and context-dependent. The Veracity Agent assigns a *NEI* verdict with moderate confidence, citing the lack of consensus across retrieved studies. The dashboard displays the verdict, evidence passages with direct links, and an exportable PDF report.

- **Example 2: Complex Multi-Claim News Article**
  A news article states: *"mRNA vaccines alter human DNA and have been linked to a surge in myocarditis deaths in athletes"*. The Decomposer Agent splits this into two sub-claims. The first is *Refuted* by multiple full-text molecular biology papers retrieved from PMC. The second receives a *NEI* label due to limited epidemiological evidence at the time of query. The Coordinator Agent aggregates these into a final *Refuted* verdict for the composite claim, and the dashboard visualizes the agent trace and per-sub-claim breakdown.

Design the dashboard so that the full system, from claim submission to sourced verdict, is accessible and interpretable for a non-expert user.

3.2 Managed Data

The system processes and manages the following types of data:

- *Input Claims*: free-text biomedical assertions, optionally extracted from URLs or documents.
- *Retrieved Evidence*: passages from PubMed abstracts, PMC full texts, and/or structured knowledge bases, with source metadata (PMID, DOI, publication date, journal).
- *Agent Intermediate Outputs*: sub-claim lists, retrieval queries, chain-of-thought justifications.
- *Structured Verdicts*: JSON objects containing the final label, confidence score, supporting evidence references, and agent execution log.
- *Historical Claim Database*: previously verified claims, queryable for analytics and trend visualization.

Benchmark datasets such as HealthFC, SciFact, and BioASQ-7b may be used to evaluate and demonstrate system performance quantitatively.

4 Related Work and Starting Points

Students are encouraged to study and build upon the following works:

- **CER-Demo** (Combating Biomedical Misinformation through Multi-modal Claim Detection and Evidence-based Verification) [1]: a multi-stage LLM-based biomedical fact-checking system integrating claim detection, PubMed evidence retrieval, and veracity assessment, with support for text, URL, and video inputs. Source code and demo available on GitHub[10].
- **CER** (Combining Evidence and Reasoning) [2]: a pipeline combining scientific evidence retrieval from PubMed, LLM-based reasoning (Mixtral-8x22B), and a binary classifier (DeBERTa/PubMedBERT), evaluated on HealthFC, BioASQ-7b, and SciFact. Source code on GitHub[11].
- **LoCal** [3]: a multi-agent fact-checking framework with decomposing, reasoning, and evaluating agents designed to handle logical and causal consistency.
- **SciClaims** [4]: an end-to-end generative system for biomedical claim extraction and verification using Llama3 and Qwen2.5 models.
- The survey on **RAG in Biomedicine** [5] provides a comprehensive overview of retrieval strategies (sparse, dense, hybrid) and LLM integration patterns relevant to this project.
- **FactAgent** [6]: a four-agent system for automated fact-checking (Input Ingestion, Query Generation, Evidence Retrieval, and Verdict Prediction agents) orchestrated by a dedicated Coordinator, evaluated on FEVEROUS, HOVER, and SciFact benchmarks with a 12.3% relative F1 improvement over baselines. The architecture and agent role decomposition offer a concrete blueprint directly transferable to the biomedical domain. Source code is available on GitHub[12].
- **PubMed and Beyond** [7]: a review of over 30 biomedical literature search tools, providing an essential map of evidence sources beyond PubMed abstracts.

5 Tools and Technologies

- **LLM Frameworks**: Hugging Face Transformers for open-source models (LLaMA-3, Mistral, Qwen, Gemma, BioMistral, etc.); GGUF-quantised models via `llama.cpp` or Ollama for local inference.

[10] `https://github.com/PRAISELab-PicusLab/CER-Fact-Checking`
[11] `https://github.com/PRAISELab-PicusLab/CER`
[12] `https://github.com/HySonLab/FactAgent`

- **Agent Orchestration**: AutoGen, LangGraph, CrewAI, or custom multi-agent implementations.
- **Retrieval**: BM25 (via BM25S or Elasticsearch), FAISS with SBERT/ BioBERT/ MedCPT embeddings; PubMed API (NCBI Entrez), PMC OA API, Europe PMC REST API.
- **NLP and Classification**: Hugging Face `pipeline` for zero-shot and fine-tuned classifiers; PubMedBERT, DeBERTa-v3, BioELECTRA.
- **Database**: MongoDB or ChromaDB / Weaviate for vector storage and provenance tracking.
- **Dashboard**: Streamlit, Dash, or Gradio.
- **PDF Report Generation** (optional): ReportLab.
- **Benchmark Datasets**: HealthFC, SciFact[13], BioASQ[14].

6 Project Evaluation

The project will be evaluated by a committee, which will assign a score from 0 to 5 to various aspects of the methodology and final dashboard. The evaluation of the methodology may vary based on the oral discussion and will be communicated only after the discussion.

6.1 Dashboard Evaluation

- **Usability**: The interface should allow a non-expert user to submit claims, inspect verdicts, and trace evidence with minimal friction.
- **Transparency**: The agent execution trace and evidence provenance must be visible and inspectable from the dashboard.
- **Accuracy of Displayed Information**: Verdicts and evidence passages must be correctly linked and clinically relevant.
- **Efficiency**: The system should return a verdict within a reasonable time, and the dashboard should allow quick navigation of historical claims.
- **Customization**: Users should be able to filter by verdict type, date, source, or topic; confidence thresholds should be adjustable.
- **Report Generation** (optional): PDF export should produce a well-structured, readable document including claim, verdict, evidence passages with citations, and confidence score.

[13] `https://github.com/allenai/scifact`
[14] `http://bioasq.org/`

6.2 Methodology Evaluation

- **Evidence Depth**: Quality and breadth of the evidence corpus; demonstrable advantage of going beyond PubMed abstracts (e.g., ablation study comparing abstract-only vs. full-text retrieval).
- **Multi-Agent Design**: Clarity of agent role separation, quality of inter-agent communication, and robustness of the orchestration logic.
- **Veracity Prediction Performance**: Quantitative evaluation on at least one benchmark dataset (HealthFC, SciFact, or BioASQ), reporting Precision, Recall, and F1.
- **Reasoning Quality**: Interpretability and faithfulness of the chain-of-thought justifications generated by the Reasoning Agent.
- **Scalability**: The ability of the system to handle a growing claim database and multiple concurrent verification requests.
- **Reproducibility and Open Science**: Use of open-source models; availability of code, configuration, and instructions to reproduce results.

References

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[6] Trinh, T., Nguyen, M., and Hy, T.-S. (2025). *Towards Robust Fact-Checking: A Multi-Agent System with Advanced Evidence Retrieval*. arXiv preprint arXiv:2506.17878. https://arxiv.org/abs/2506.17878

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